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ON QUASI-DEGENERACIES IN PLATE VIBRATION PROBLEMS

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Summary—In plotting the variation of frequencies with geometric parameters such as side ratio, skew angle, thickness taper, etc. in detailed studies of the vibration characteristics of plates, situations are encountered such as crossing of the frequency curves or the tendency of these curves to come close together and veer away from each other. These have been generally referred to as "frequency crossings" and "transitions" respectively. The latter may preferably be referred to as "quasi-degeneracies". In the literature there appears to be some ambiguity in the analysis and interpretation of these features. In this paper, a clarification of some of these questions as regards rectangular and skew plates is presented by making use of concepts from physics dealing with molecular vibrations.

NOTATION

a, b	dimensions of the plate, see Fig. 1
D	plate stiffness, $Eh^3/12(1-\nu^2)$
H, H'	differential operators, see equations (2) and (3)
H_p	perturbation operator, see equation (3)
h'_{mn}	integral defined in equation (8)
i	integer
k, k'	eigenvalue parameters, see equations (1, 3)
k_1, k_2	two close eigenvalues of the unperturbed problem, equation (1)
k'_1, k'_2	two close eigenvalues of the perturbed problem, equation (3)
m, n, r, s	integers
S	domain of the plate
$S_1, S_2, \ldots, S_n$	symmetry groups in a system
W_m	mth eigenfunction
W, W'	eigenfunctions, see equations (1) and (3)
$W_m^{(n)}$	mth eigenfunction belonging to the symmetry group S_n
ξ, η	non-dimensional co-ordinates
λ	side ratio, a/b
ϕ	notation for function $\phi(x, y)$
ψ	skew angle

INTRODUCTION

IN STUDIES of the vibration characteristics of plates with different geometries and boundary conditions, etc., the variation of frequencies of the modes are plotted as the geometric parameters are varied. It has been observed [see, for example, ref. (1)] that the frequency curves appear either to cross each other or just come close together and veer away from each other. The former are known as "frequency crossings" and the latter have been referred to as "transition

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points". It appears preferable, however, to call these "quasi-degeneracies". In the case of plate configurations without any symmetry, considerable ambiguity seems to exist in ascertaining which of the two is correct. A detailed analysis is, obviously, necessary in all such cases of frequency crossings and quasi-degeneracies for a complete understanding of the vibration characteristics of plates. In the vicinity of quasi-degeneracies, the nodal patterns are known to undergo rapid changes. In flutter problems, such rapid variations of mode shapes are quite important as they are likely to result in significant changes in the flutter characteristics.

Warburton[2] first discussed in detail the rapid changes of nodal patterns associated with a quasi-degeneracy in the case of clamped rectangular plates in a comprehensive study of the vibration characteristics of rectangular plates. Claassen and Thorne[3,4] gave the frequency parameters computed at very close intervals of side ratio in the neighbourhood of frequency crossings, and transitions (quasi-degeneracies) in the case of cantilever and clamped rectangular plates. These results show that the frequency curves belonging to the same symmetry group do not cross each other while the frequency curves, each belonging to a different symmetry group, can cross. Webster[5] also discusses these features in his paper on the vibration of rectangular curved panels. The quasi-degeneracies were also observed in refs. (6) and (7) dealing with vibration characteristics of skew membranes and plates and they are even more manifest in our recent investigations on the vibration problems of skew plates with different edge conditions.[8] Quasi-degeneracies are known to occur quite frequently among the frequency curves of plates with in-plane stresses and, in particular, under shear.[9,10] More recently, Fauconneau and Marangoni[11] also report similar observations in their studies of stressed, simply supported rectangular plates. Investigations of this nature with respect to eigenvalues of Schrödinger's wave equation are well known, and are available in standard textbooks on atomic and molecular spectroscopy.[12] When the edge conditions of the plate are diverse, there is no symmetry of the plate as regards the boundary conditions, and the feature of quasi-degeneracy only is seen as frequency crossings *per se* are ruled out.

In this paper, we attempt a detailed study of the frequency crossings and quasi-degeneracies of frequency curves due to variations of the geometric parameters like side ratio and skew angle. It is shown, by an adaptation of perturbation theory for adjacent energy levels in the calculation of molecular spectra,[13] that, for configurations where symmetry exists, the frequency curves of modes of the same symmetry group cannot cross while those of different symmetry groups can cross, and where there is no symmetry, no such crossings can occur. Also, a closer study of the variation of frequencies and nodal patterns in the region of quasi-degeneracies is attempted for a few representative cases. The changes in the relative proportions of the different terms in the eigenvector and the resulting changes in the nodal patterns in the neighbourhood of these quasi-degeneracies are illustrated. The extension of this analysis to include the variations of eigenvalues with respect to thickness, mass distribution and in-plane loads is straightforward.

MATHEMATICAL TREATMENT

The eigenvalue problem of plate vibration within the framework of classical, small deflection, isotropic, thin plate theory may be represented in operator notation, as,

$$HW = kW, \tag{1}$$

where H is a self-adjoint partial differential operator,[14] k the eigenvalue and W the eigenfunction of the problem. In terms of oblique coordinates (see Fig. 1)

$$H = \partial^4/\partial\xi^4 + \lambda^4\,\partial^4/\partial\eta^4 + 2(1+2\sin^2\psi)\,\partial^4/\partial\xi^2\,\partial\eta^2$$
$$-4\lambda\sin\psi(\partial^4/\partial\xi^3\,\partial\eta + \lambda^4\,\partial^4/\partial\xi\,\partial\eta^3). \tag{2}$$

Let k_1, k_2 be two close eigenvalues and W_1, W_2 be the corresponding eigenfunctions. It is required to ascertain the nature of variation of the eigenvalues k_1 and k_2 when a small perturbation ψ_p is given to the skew angle. It may be assumed that the perturbation is so small that one can ignore the influence of modes with frequencies far separated from the pair in question.

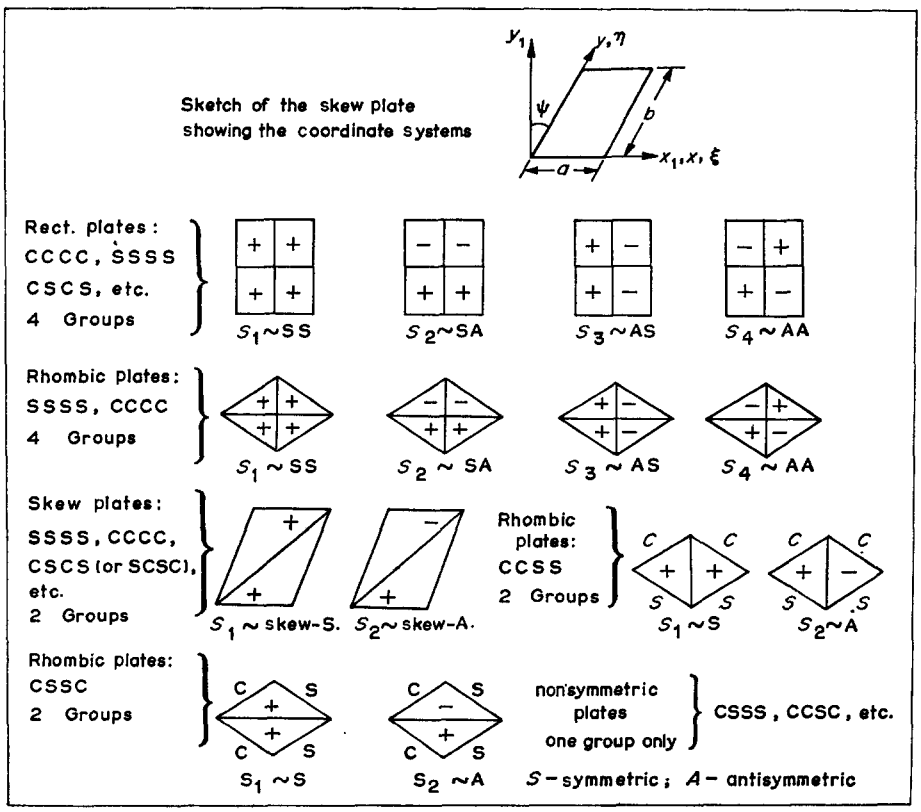

FIG. 1. Symmetry groups in different plate configurations.

The perturbed governing differential equation may be written formally as

$$H'W' = (H+H_p)\,W' = k'W', \tag{3}$$

where k' and W' are now the eigenvalue and eigenfunction, respectively, of the perturbed problem. In the case of a skew plate, for small perturbations ψ_p in skew angle ψ, operators

H' and H_p are as follows:

$$H' = \partial^4/\partial\xi^4 + \lambda^4\,\partial^4/\partial\eta^4 + 2[1+2\sin^2(\psi+\psi_p)]\,\partial^4/\partial\xi^2\,\partial\eta^2 - 4\lambda\sin(\psi+\psi_p)\,(\partial^4/\partial\xi^3\,\partial\eta + \lambda^4\,\partial^4/\partial\xi\,\partial\eta^3),$$

$$H_p = 4\psi_p\sin 2\psi\,\partial^4/\partial\xi^2\,\partial\eta^2 - 4\lambda\psi_p\cos\psi(\partial^4/\partial\xi^3\,\partial\eta + \lambda^4\,\partial^4/\partial\xi\,\partial\eta^3).$$

It is easily verified that H_p is also a self-adjoint operator. The solution of the perturbed problem, equation (3), can be expressed as a linear combination of the unperturbed solutions[13], i.e.

$$W' = C_1\,W_1 + C_2\,W_2. \tag{4}$$

Now, using equation (4), equation (3) can be written as

$$(H+H_p)\,(C_1\,W_1 + C_2\,W_2) = k'(C_1\,W_1 + C_2\,W_2), \tag{5}$$

which gives, after rearranging,

$$(k'-k_1)\,C_1\,W_1 + (k'-k_2)\,C_2\,W_2 = H_p(C_1\,W_1 + C_2\,W_2). \tag{6}$$

Multiplying equation (6) by W_1 and W_2 in succession, and integrating over the domain of the plate, one gets

$$\left.\begin{aligned}(k'-k_1)\,C_1 &= h'_{11}\,C_1 + h'_{12}\,C_2,\\ (k'-k_2)\,C_2 &= h'_{21}\,C_1 + h'_{22}\,C_2,\end{aligned}\right\} \tag{7}$$

where

$$h'_{mn} = \int_S\int W_m\,H_p\,W_n\,\mathrm{d}s. \tag{8}$$

As H_p is self-adjoint, $h'_{nm} = h'_{mn}$. For the pair of equations in equation (7) to have non-trivial solution, the determinant

$$\begin{vmatrix} k'-k_1-h'_{11} & -h'_{12} \\ h'_{21} & k'-k_2-h'_{22}\end{vmatrix} = 0, \tag{9}$$

giving

$$(k'-k_1-h'_{11})\,(k'-k_2-h'_{22}) - h'_{12}\,h'_{21} = 0,$$

or

$$k'_{1,2} = \tfrac{1}{2}[(k_1+h'_{11}) + (k_2+h'_{22})] \pm \tfrac{1}{2}\{[(k_1+h'_{11}) - (k_2+h'_{22})]^2 + 4(h'_{12})^2\}^{\frac{1}{2}}. \tag{10}$$

From equation (10), it is obvious that crossing or non-crossing of the curves representing the variation of eigenvalues k'_1 and k'_2 with ψ is governed by the value assumed by h'_{12}. We shall examine different cases in which W_1 and W_2 have different symmetry properties to study this behaviour of the frequency curves.

1. *Problems with symmetry*

In problems where symmetry exists, the eigenfunctions $W_1, W_2, W_3, \ldots$ can be grouped into different symmetry groups, $S_1, S_2, S_3, \ldots, S_n$ (the actual number of symmetry groups, obviously, depends upon the order of symmetry in the problem). Denoting $W_r^{(n)}$ as the rth eigenfunction belonging to the symmetry group S_n, we can write

$$H_p\,W_2^{(n)} = \phi^{(n)}. \tag{11}$$

As H_p is a symmetric operator, the symmetry characteristics of the function ϕ will be the same as that of W_2. It is readily seen that a series expansion of $\phi^{(n)}$ in terms of W_n's will contain only terms of S_n symmetry. In other words,

$$\phi^{(n)} = \sum_{r=1}^{\infty} a_r\,W_r^{(n)}. \tag{12}$$

The value of the integral h'_{12}, which determines whether the frequency curves corresponding to k'_1 and k'_2 would cross, depends on the symmetry properties of W_1 and W_2.

(a) *Cases in which W_1 and W_2 belong to the same symmetry group (S_m)*

In this case,

$$h'_{12} = \int_S \int W_1^{(m)} H_p W_2^{(m)} \, ds$$

$$= \int_S \int W_1^{(m)} \sum_{r=1}^{\infty} a_r W_r^{(m)} \, ds = a_1. \tag{13}$$

Hence from equation (10) and equation (13) it is seen that whenever W_1 and W_2 belong to the same symmetry group, the value of h'_{12} does not, in general, vanish. Consequently, the frequency curves corresponding to k'_1 and k'_2 cannot cross each other. However, the possibility of the two frequency curves touching cannot be totally ruled out. The two frequencies k'_1 and k'_2 would become equal if the perturbation is such that [see, equation (10)]

$$[(k_1+h'_{11})-(k_2+h'_{22})]^2+4(h'_{12})^2 \equiv 0. \tag{14}$$

This means that both

$$h'_{12} \equiv 0 \tag{15}$$

and

$$(k_1+h'_{11})-(k_2+h'_{22}) \equiv 0. \tag{16}$$

Clearly, these two conditions cannot be satisfied simultaneously by varying only one parameter (in this case λ or ψ), and hence the non-crossing rule of frequency curves of the same symmetry group [see ref. (15)].

The non-crossing rule dealt with in this section has one case which is an exception and this is discussed in section 1(c).

(b) *Cases in which W_1 and W_2 belong to different symmetry groups (say S_n and S_m)*

In this case,

$$h'_{12} = \int_S \int W_1^{(n)} H_p W_2^{(m)} \, ds = \int_S \int W_1^{(n)} \sum_{r=1}^{\infty} a_r W_r^{(m)} \, ds = 0. \tag{17}$$

From equation (10), the two frequencies can now be written as

$$k'_1 = k_1+h'_{11}, \quad k'_2 = k_2+h'_{22}. \tag{18}$$

It is clear from equation (18) that the variation of frequencies k'_1 and k'_2 is independent of each other. In other words, each curve can be drawn virtually ignoring the presence of the other and the two curves can cross each other. The identity of the curves beyond the crossing always exists by virtue of the fact that they belong to different symmetry groups. So, whenever W_1 and W_2 belong to different symmetry groups, the corresponding frequency curves can simply cross each other without mutual interaction.

(c) *The exception to the non-crossing rule*

The exception to the non-crossing rule occurs when

$$H_p W_2 = c W_2, \tag{19}$$

c being a constant. In such a case

$$h'_{12} = \int_S \int W_1 H_p W_2 \, ds = c \int_S \int W_1 W_2 \, ds = 0, \tag{20}$$

from the orthogonality of normal modes.

A significant example of this exceptional case is the uniform, simply supported rectangular plate or membrane for which it is well known that the modes are given by products of sine functions,

$$\left.\begin{aligned} W_1 &= A \sin m\pi\xi \sin n\pi\eta \\ \text{and} \quad W_2 &= B \sin r\pi\xi \sin s\pi\eta. \end{aligned}\right\} \tag{21}$$

When m and r and n and s are *both* either even or odd, W_1 and W_2 belong to the same symmetry group. Here $h'_{12} = 0$ irrespective of the values of m, n, r and s provided $m \neq r$ and $n \neq s$. Hence in the case of uniform, simply supported rectangular plates or membranes, two frequency curves belonging to the same symmetry group can also cross.

2. *Problems with no symmetry*

In this case, the solutions $W_1, W_2, W_3, \ldots$ of the unperturbed problem, equation (1), are unsymmetric. So $\phi = H_p W_2$ will also be unsymmetric. Again, expanding ϕ in terms of the set of functions $W_1, W_2, W_3, \ldots$

$$\phi = \sum_{r=1}^{\infty} a_r W_r, \tag{22}$$

giving

$$h'_{12} = \int_S \int W_1 H_p W_2 \, \mathrm{d}s = \int_S \int W_1 \sum_{r=1}^{\infty} a_r W_r \, \mathrm{d}s = a_1. \tag{23}$$

This case can be seen to be similar to the case in which W_1 and W_2 belong to the same symmetry group. The value of h'_{12} is not generally equal to zero and the two conditions given by equation (16) and equation (17) cannot once again be satisfied simultaneously by variation of a single parameter. Thus, in the absence of any symmetry in the problem, no two frequency curves would cross each other.

3. *Some general remarks*

It may be useful at this stage to emphasize some of the differences between a frequency crossing and a quasi-degeneracy. As shown earlier, in the neighbourhood of a crossing, the value of h'_{12} is zero at every point and the modes of these frequencies do not interact with each other. In the case of a quasi-degeneracy, where frequency curves come close together and veer away with or without touching each other, the value of h'_{12} is everywhere non-zero except at the point of contact (if it exists) and involves linear combination and significant interaction between the two constituent modes.

It appears instructive to draw attention to the dependence of the frequencies of oscillation, of a two degree-of-freedom coupled system, on the magnitude of the coupling. A plot of the two natural frequencies against the ratio of uncoupled frequencies of the system along the abscissa and with the magnitude of coupling as the parameter [see, for example, Fig. 184, ref. (16)] has features similar to crossings and quasi-degeneracies of pairs of frequency curves. When the coupling term is zero, the system reduces to two separate uncoupled single degree-of-freedom systems and the frequency curves cross each other. When the coupling term is not zero, significant interaction between the two frequencies is present and the curves do not cross but only come close together and veer away from each other. The similarity of the role of the coupling term in this standard coupled oscillator problem with h'_{12} of our present discussion is very vivid indeed.

SYMMETRIES OF PLATE CONFIGURATIONS

Before discussing the results depicting the features of frequency crossings and quasi-degeneracies in the case of rectangular and skew plates, it would be useful to consider the symmetry classification of modes of different plate configurations (also see Fig. 1).

(a) *Rectangular plates*

The symmetry groups of the modes of vibration, obviously, depend on the boundary conditions. Thus, in the case of rectangular plates with their edges supported symmetrically (CCCC, SSSS, FFFF, CSCS, CFCF or SFSF plates)* there are two lines of symmetry and the modes can be grouped into four symmetry groups with medians as the axes of symmetry. They are

$S_1 \sim$ Symmetric–Symmetric (SS),

$S_2 \sim$ Symmetric–Antisymmetric (SA),

$S_3 \sim$ Antisymmetric–Symmetric (AS),

$S_4 \sim$ Antisymmetric–Antisymmetric (AA).

* The edge conditions have been described using a combination of the single letters C, S and F standing for clamped, simply supported and free, and going round the plate clockwise starting with the left-hand edge.

For CCSC, CSSS, CFFF, SFFF and CFSF rectangular plates there is only one axis of symmetry, namely, one of the medians, and the modes can be grouped into two symmetry groups, symmetric and antisymmetric about this axis. In the case of CSSC, CFFC and SFFS square plates also there is only one axis of symmetry, a diagonal, with two symmetry groups of modes.

Rectangular plates with CSSC, CFFC and SFFS boundary conditions and square plates with CSSF, CCSF and CFFS boundary conditions are examples of unsymmetric cases.

(b) *Skew plates*

Among skew plates, the modes of rhombic plates with CCCC, SSSS and FFFF boundary conditions can be divided into the following four symmetry groups, with diagonals as the axes of symmetry:

$$S_1 \sim \text{Symmetric–Symmetric},$$

$$S_2 \sim \text{Symmetric–Antisymmetric},$$

$$S_3 \sim \text{Antisymmetric–Symmetric},$$

$$S_4 \sim \text{Antisymmetric–Antisymmetric}.$$

This is true for rhombic membranes also when all the sides are fixed.

CCCC, SSSS, FFFF skew plates ($a/b \neq 1$) and CSCS, CFCF and SFSF rhombic and skew plates belong to the category of plates with two symmetry groups. The two groups are

$$S_1 \sim \text{Skew Symmetric},$$

$$S_2 \sim \text{Skew Antisymmetric}.$$

Again, CSSC, CFFC and SFFS rhombic plates have symmetry about one of their diagonals. The two symmetry groups in this case are

$$S_1 \sim \text{Symmetric},$$

$$S_2 \sim \text{Antisymmetric}.$$

Combinations of boundary conditions other than those discussed above, i.e. CCSC, SFFF, etc., in rhombic plates, and CSSC etc., in skew plates, are examples of unsymmetric cases.

ANALYSIS OF RESULTS

In Fig. 2, the variation of the frequency parameter of a clamped rectangular plate with side ratio, given in ref. (4), is replotted using the numerical values reported in ref. (1). It is seen that the third and fourth modes of the SS group undergo quasi-degeneracy at about $a/b = 0{\cdot}62$.* Similarly the second and third modes and the third and the fourth modes of the SA group also undergo quasi-degeneracies at around $a/b = 0{\cdot}83$ and $a/b = 0{\cdot}54$, respectively. It is seen that there are several frequency crossings between modes belonging to different symmetry groups.

Fig. 3 shows the variation of frequency with side ratio for simply supported rectangular plates which constitute the exceptional case as mentioned earlier. These curves show that in addition to crossing of frequency curves belonging to different symmetry groups (S_1, S_2, S_3 and S_4) frequency curves belonging to the same symmetry group also cross. Thus, modes 1/4† and 3/2 belonging to the SA symmetry group cross at $a/b = (2/3)^{\frac{1}{2}}$ and modes 1/5 and 3/3 of the SS group at $a/b = (1/2)^{\frac{1}{2}}$. The reason for this exceptional behaviour in this case has been explained already through equations (10), (19)–(21).

* The second and third modes also of the SS group undergo quasi-degeneracy at $a/b = 1$ but as a/b is limited to 1 only in the figure this is not seen completely in Fig. 2.

† A mode with m half-waves in the ξ-direction and n half-waves in the η-direction is generally denoted as an m/n mode.

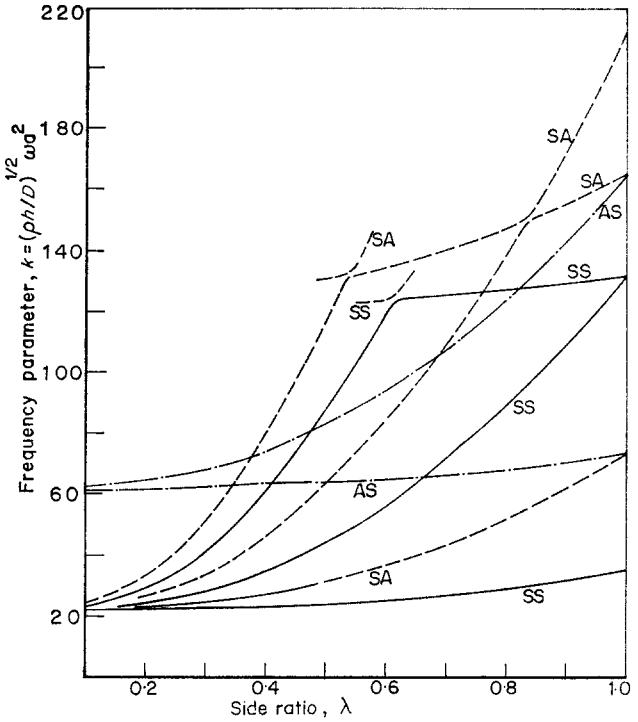

FIG. 2. Frequency parameters of CCCC rectangular plate [ref. (1)].

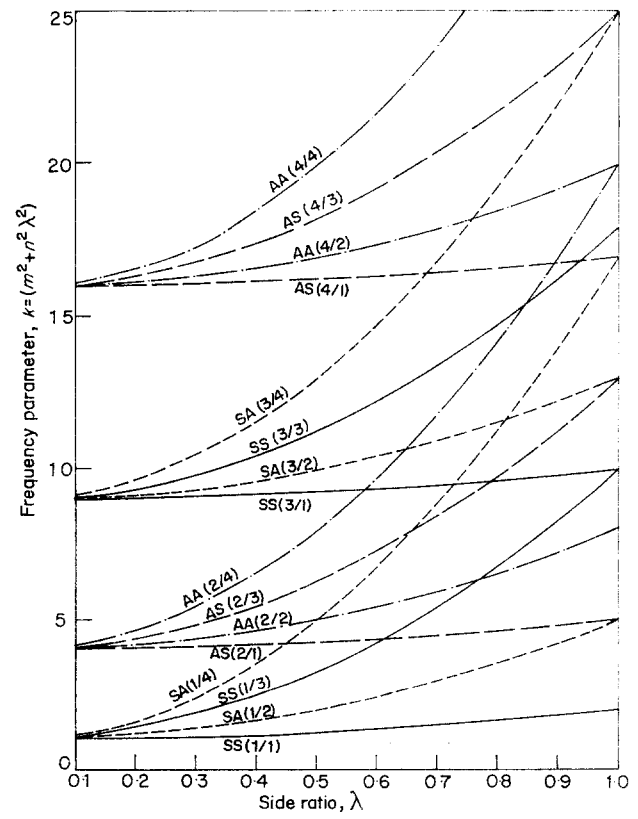

FIG. 3. Frequency parameters of SSSS rectangular plate.

The details of crossing and quasi-degeneracies occurring in SSSS and CCCC skew plates may be seen from the results presented and discussed in refs. (6) and (7). The details of such features for unsymmetric configurations, i.e. skew plates with different combination of boundary conditions, are discussed in ref. (8), wherein the calculations of frequencies and nodal patterns is based on a 36-term Rayleigh–Ritz solution using the products of beam characteristic functions as the approximating functions. More explicitly, the deflection shape is represented as

$$W(\xi, \eta) = \sum_{m}^{6} \sum_{n}^{6} C_{mn} X_m(\xi) Y_n(\eta),$$

where $X_m(\xi)$ and $Y_n(\eta)$ are the appropriate beam characteristic functions. For a CCSC plate, for example, $X_m(\xi)$ and $Y_n(\eta)$ are the clamped–supported and clamped–clamped beam functions. The variation of frequencies with skew angle for one typical unsymmetric configuration, CCSC, is shown in Fig. 4 along with the variation of frequencies of the skew symmetric configuration SCSC (which is the same as CSCS). The curves in Fig. 4 show how

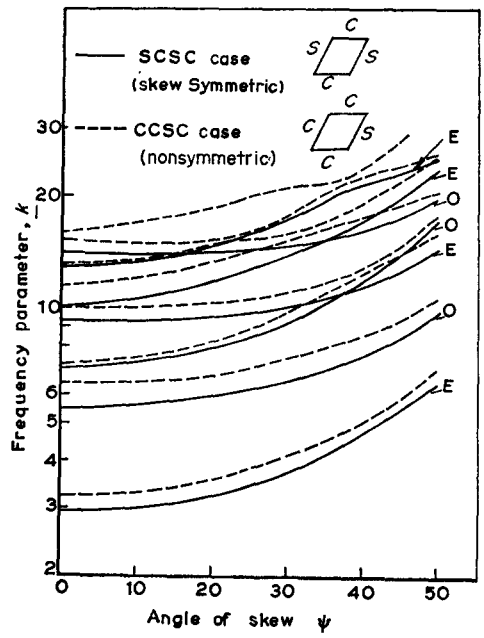

FIG. 4. Effect of introducing asymmetries.

the introduction of asymmetries, in this case by altering the boundary conditions, changes the frequency crossings to quasi-degeneracies. The figure shows that in a CCSC rhombic plate, quasi-degeneracies occur between the third and fourth modes at about 40°, between the fifth and sixth at about 33° and between the sixth and seventh at about 21°. Frequency crossings are totally absent in this unsymmetric case. An analogous situation in the field of molecular vibrations is seen from Fig. 6 of ref. (17), and an examination of this figure would indicate the striking similarities.

A closer study of a typical quasi-degeneracy, namely, that between the third and fourth modes, is shown in Fig. 5. Frequency parameters and nodal patterns are computed at close intervals of the skew angle in the neighbourhood of the quasi-degeneracy and are plotted. The nodal patterns plotted in the figure show rapid changes in the mode shapes in the neighbourhood of such quasi-degeneracies. The rapid variation of these two modes is also examined in Fig. 6 by plotting the percentage predominance of different terms in the eigenvectors of the two modes. If $\{C_{mn}^{(i)}\}$ is the eigenvector corresponding to the

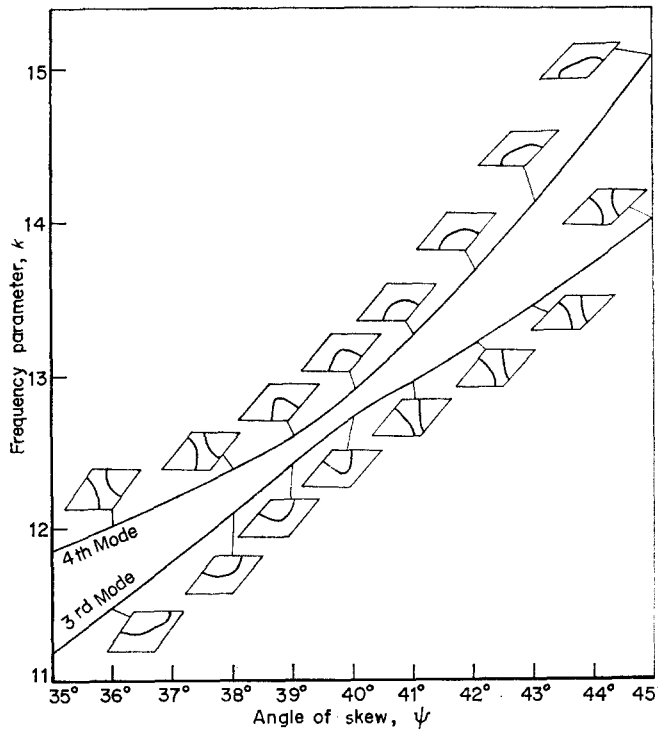

FIG. 5. Variation of frequencies and modes in the neighbourhood of quasi-degeneracy.

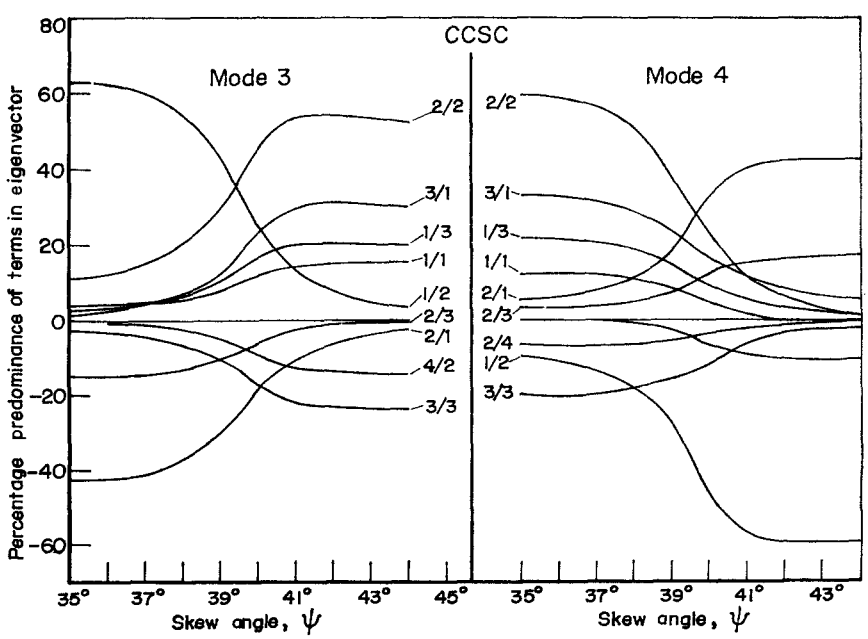

FIG. 6. Variation of percentage predominance of terms in the eigenvectors in the neighbourhood of quasi-degeneracy.

ith mode (in the present case, a 36×1 matrix), the percentage predominance of the term C_{rs} or the rsth mode (shown in the figure with the usual notation r/s), is calculated as $100\, C_{rs}/(\sum_{m=1}^{6} \sum_{n=1}^{6} C_{mn}^2)$. This plot clearly shows the changes in the relative proportions of the constituent modes in the neighbourhood of a quasi-degeneracy. While predominance of different terms changes very rapidly in this region, the contribution of higher terms generally remains very small. Also, it is clearly seen that, in the eigenvector of the third mode, the skew antisymmetric terms are predominant compared to skew symmetric terms before the quasi-degeneracy. The skew antisymmetric terms rapidly decrease in magnitude during the transition while skew symmetric terms correspondingly increase. The reverse tendency is seen in the case of the fourth mode. We recall that Fig. 5 had shown that while the nodal patterns of the third and fourth modes are largely skew antisymmetric and skew symmetric respectively before the quasi-degeneracy, this nature gets interchanged after this point. In the immediate vicinity of quasi-degeneracy the nodal patterns do not possess any such specific affiliations. These features associated with quasi-degeneracy are found to occur also in the case of CSSS rhombic plates,[8] plates with a thickness taper[18, 19] and also in anisotropic plates.[20, 21]

One implication of the above discussions is that the assumptions of simple mode shapes sometimes made in approximate solutions of vibration problems would not be valid in the vicinity of quasi-degeneracies and might result in an error. In fact, the recent discussion of Webster and Warburton[22] on the paper by Szechenyi[23] dealing with determination of natural frequencies of stiffened and curved panels eloquently directs attention to this question. The significant interaction between the neighbouring modes in the vicinity of a quasi-degeneracy is responsible for this limitation.

CONCLUSIONS

A study of the variations of frequencies with geometric parameters like skew angle and side ratio is attempted with special emphasis on clarifying features with regard to "frequency crossings" and "quasi-degeneracies" in rectangular and skew plates with different edge conditions. It is shown analytically, using concepts from molecular vibrations, that the frequency curves belonging to different symmetry groups of plates can cross each other and that those belonging to the same symmetry group cannot do so. In the case of unsymmetric plate configurations, the frequency crossings are totally absent and only quasi-degeneracies occur. These features are illustrated with some typical examples from rectangular and skew plates.

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REFERENCES

1. A. W. Leissa, *Vibration of Plates*, NASA SP-160 (1969).
2. G. B. Warburton, *Proc. Inst. Mech. Engng* **168**, 371 (1954).
3. R. W. Claassen and C. J. Thorne, *J. Aerospace Sci.* **29**, 1300 (1962).
4. R. W. Claassen and C. J. Thorne, *J. appl. Mech.* **28**, 304 (1961).
5. J. J. Webster, *Int. J. mech. Sci.* **10**, 571 (1968).
6. S. Durvasula, *J. acoust. Soc. Am.* **44**, 1636 (1968).
7. S. Durvasula, Report No. AE 219 S, Department of Aeronautical Engineering, Indian Institute of Science, Bangalore (1969); also in *AIAA J.* **7**, 1164 (1969).
8. P. S. Nair and S. Durvasula, *J. Sound Vibrat.* **25**, 1 (1973).
9. J. G. Eisley and G. Luessen, *AIAA J.* **1**, 620 (1963).
10. S. Durvasula and V. Sundara Ramaiah, *Sixteenth Cong. of Theoretical and Applied Mechanics*, March 29–31, M.N.R. Engng College, Allahabad (1972).
11. G. Fauconneau and R. D. Marangoni, *Int. J. Solids Struct.* **7**, 743 (1971).

12. G. HERZBERG, *Infrared and Raman Spectra of Polyatomic Molecules*. Van Nostrand, New York (1946).
13. A. B. MIGDAL and U. KRAINOV, *Approximation Methods in Quantum Mechanics*. Benjamin, New York (1969).
14. P. M. MORSE and H. FESHBACH, *Methods of Theoretical Physics*. McGraw-Hill, New York (1953).
15. E. TELLER, *J. phys. Chem.* **XLI**, 109 (1937).
16. K. MAGNUS, *Vibrations*. Blackie, Glasgow (1965).
17. P. A. FLEURY, *J. Acoust. Soc. Am.* **49**, 1041 (1971).
18. R. PLUNKETT, *J. mech. Engng Sci.* **5**, 146 (1963).
19. I. CHOPRA and S. DURVASULA, *Int. J. mech. Sci.* **13**, 935 (1971).
20. N. J. HUFFINGTON, JR., *J. appl. Mech.* **28**, 459 (1961).
21. P. S. NAIR and S. DURVASULA, *J. Acoust. Soc. Am.*, to be published.
22. J. J. WEBSTER and G. B. WARBURTON, *J. Sound Vibrat.* **18**, 139 (1971).
23. E. Szechenyi, *J. Sound Vibrat.* **14**, 401 (1971).